Global and Token-Level Crystallographic Text for Crystal Property Prediction: Complementarity and Prediction Averaging

Huaijuan Zang[a], Yunfan Peng[a,*], Chong Zhao[b], Fan Yang[a], Feng Hong[c] and Liangfeng Xu[a]

[a]School of Computer and Information, Hefei University of Technology, Hefei, China
[b]Engineering Quality Education Center, Hefei University of Technology, Hefei, China
[c]Chizhou University, Chizhou, China

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ABSTRACT

Crystal property predictors can combine periodic graphs with different aggregations of the same generated crystallographic text. We examine how text aggregation affects prediction and whether averaging global and token-level predictors provides gains beyond same-view ensembling. Independently trained SFTGNN–MatBERT predictors are evaluated on six JARVIS and four Materials Project tasks using fixed data partitions and two training seeds. Their fixed 0.5/0.5 prediction mean achieves an average task-wise relative MAE reduction of approximately 5.9% compared with the Global baseline, improving over Global on all ten tasks and over both constituent predictors on nine tasks in each run. A separate matched six-task validation ablation finds no overall advantage for graph-conditioned cosine attention over CLS or masked mean pooling: their normalized macro MAEs are 1.0291, 1.0000, and 0.9979, respectively. Same-view ensemble controls further show that the cross-view combination does not outperform token-only ensembling at the normalized-macro level. Exploratory interaction variants exhibit property-dependent behavior under nonuniform validation protocols. Together, these results distinguish single-predictor aggregation performance from prediction-level complementarity and show why same-view ensemble controls are necessary when interpreting gains from graph–text prediction averaging.

1. Introduction

High-throughput electronic-structure databases and learned property predictors support the screening of crystalline materials across large chemical and structural spaces. Resources such as the Materials Project, JARVIS, AFLOW, OQMD, and Matminer provide data and tools for this workflow [1–5]. Crystal graph neural networks encode atomic species, periodic connectivity, and geometry to estimate properties directly from structures. CGCNN established a widely used message-passing approach [6], while subsequent models incorporate richer geometric information. For graph–text prediction, a complementary question is how to use a crystallographic description derived from the same structure.

Robocrystallographer expresses coordination environments, connectivity, symmetry, and structural motifs in natural language [7]. This text is a derived structural view, not an independent experimental observation. A materials language encoder can represent the input sequence through a document-level CLS state or through contextual token states. Here, "global" refers to the CLS representation of the tokenized input retained within the sequence-length limit; "token-level" refers to aggregation over its eligible contextual token states. Both representations can contain sequence-wide context. Their distinction concerns how that context is aggregated, rather than a strict separation between global and local physical information.

Existing studies establish the feasibility of combining crystal graphs and language representations. CrysMMNet uses document-level language features, LatticeLingo examines descriptions at different structural scopes, and CAST uses fine-grained structure–text cross-attention [8–10]. These approaches motivate access to richer textual features, but greater representational detail does not by itself establish better property prediction. Moreover, an improvement obtained by combining two predictors can arise from ordinary prediction averaging as well as from differences in their text inputs or aggregation mechanisms. These possibilities require separate comparisons.

*Corresponding author: Yunfan Peng
zanghj@hfut.edu.cn (H. Zang); luoai4430@gmail.com (Y. Peng); zhaochong@hfut.edu.cn (C. Zhao); 2021800201@hfut.edu.cn (F. Yang); hongfeng@czu.edu.cn (F. Hong); xulfcjn@hfut.edu.cn (L. Xu)
ORCID(s):

We therefore ask two questions. First, how does aggregation of frozen MatBERT representations affect prediction when the downstream head and training objective are held fixed? Second, does averaging independently trained global and token-level predictors improve on both individual predictors and same-view ensembles? The first question concerns single-predictor performance; the second concerns prediction-level complementarity. Evidence for one does not automatically answer the other.

We address the ensemble question with Global and Token pathways built on independent SFTGNN encoders. The Global pathway uses graph–CLS regression, whereas the Token pathway uses graph-conditioned cosine attention and a residual prediction head. Because these pathways also differ in their heads and objectives, their comparison evaluates complete predictor configurations. A separate matched validation ablation compares CLS, masked mean pooling, and cosine attention under a common head and loss. Same-view ensembles and sample-level error diagnostics provide controls for interpreting the main prediction averages. Additional message-level, latent-level, and shared-backbone variants are retained as exploratory observations rather than a controlled search for the best fusion location.

The contributions are threefold. First, we document the behavior of fixed cross-view prediction averaging across ten property tasks and two training seeds, with sample alignment and validation-based checkpoint selection. Second, we compare this behavior with same-view ensembles and a matched six-task representation ablation, separating evidence about aggregation accuracy from evidence about ensemble gains. Third, we report sample-level error diagnostics and property-dependent interaction results that define the scope of the empirical findings. The aim is to establish which claims the comparisons support, rather than to assume that finer textual granularity or more direct interaction is inherently preferable.

2. Related Work

2.1. Crystal graph neural networks

Crystal GNNs can be organized by the geometric information emphasized in message passing. Distance-based methods use local neighborhoods and radial functions. CGCNN learns edge-conditioned updates from atom and bond descriptors [6], while SchNet learns continuous filters from pair distances [11]. State-based architectures such as MEGNet jointly update edges, nodes, and global attributes, enabling broad materials and molecular prediction [12].

Directional models enrich radial descriptions with angular structure. DimeNet constructs directional messages using radial and spherical basis functions [13]; ALIGNN alternates atomistic and line-graph propagation to model bond angles explicitly [14]. Equivariant networks constrain feature transformations under rotations and can achieve high data efficiency for atomistic energies and forces; NequIP is a representative example [15]. Periodic graph transformers include Matformer and the complete geometric representation of ComFormer [16, 17]. SFTGNN introduces spherical-Fourier-enhanced crystal message passing [18]. Composition-only predictors such as Roost and CrabNet provide complementary structure-free references [19, 20]. Related foundation-model efforts in materials increasingly emphasize transferable atomistic or conversational representations, including universal interatomic potentials and multimodal materials assistants [21–24]. These approaches primarily solve geometric, atomistic, or compositional representation; they do not by themselves determine how graph information should interact with document-level and token-level crystallographic language representations.

2.2. Materials language models

Scientific language differs from general prose through dense entities, formulae, measurement conventions, and domain-specific relations. Materials text additionally contains chemical formulae, space-group symbols, coordination terminology, bonding descriptions, and names for structural prototypes or motifs. MatBERT adapts transformer language modeling to materials corpora, and MatSciBERT provides a domain scientific encoder suited to materials-related extraction and representation [25, 26]. Their contextual token states offer richer semantics than fixed word vectors, but downstream use often retains only a pooled or CLS representation.

Robocrystallographer provides a controlled interface between structure and language by generating descriptions of symmetry, coordination, and connectivity from a crystal structure [7]. This consistency is valuable for multimodal learning: differences in text arise from structural differences rather than uncontrolled author style. Nevertheless, generated descriptions inherit the heuristics and vocabulary of the description engine, so text should be understood as a complementary structural view rather than an independent ground truth.

2.3. Multimodal materials learning

Multimodal representation learning has been strongly shaped by image–text alignment models such as CLIP, ALBEF, and BLIP, which show that paired modalities can be connected through contrastive objectives, cross-modal attention, and bootstrapped caption supervision [27–29]. These models motivate a general principle: multimodal learning can benefit from both shared alignment spaces and modality-specific representations. For crystal property prediction, however, the setting differs because the text description is generated from the crystal structure rather than collected as an independent observation. The modalities are therefore partially redundant, and the main question becomes how different abstractions of the same material should be exploited.

Multimodal materials learning has combined structure, composition, images, spectra, and text for prediction or representation learning. Graph–text models are particularly attractive because both modalities can be generated for the same crystal and aligned at the sample level. CrysMMNet combines crystal-graph representations with document-level language representations, whereas LatticeLingo studies descriptions constructed at local, semiglobal, and global structural scopes [8, 9]. CAST introduces cross-attention-based structure–text interaction, while Hybrid-LLM-GNN combines graph and materials-language embeddings for cross-property prediction [10, 30]. CLICS explores graph–text contrastive learning, and MatterChat broadens materials multimodality toward conversational foundation models [24, 31]. The present study examines aggregation choices within one generated description and evaluates the errors of independently trained predictors. Its main distinction is the joint use of matched aggregation comparisons and same-view ensemble controls to interpret the benefit of cross-view prediction averaging.

2.4. Fine-grained cross-modal attention

Cross-attention is widely used to align elements of two modalities following scaled dot-product attention in the Transformer [32]. In materials prediction, a graph-level query and textual token keys provide a natural asymmetric form: the crystal asks which parts of its own description are useful. Dot-product similarity, however, combines direction and norm. When encoders and projections have different scale statistics, attention sharpness can be governed by norms instead of correspondence. Query-Key Normalization motivates normalizing both projections to prevent arbitrary softmax saturation [33]. Related normalization ideas also appear in contrastive learning, where CLIP uses normalized image and text features with a learned temperature to stabilize alignment [27]. Our token pathway adopts a normalized query–key formulation and treats entropy, effective token count, and maximum attention as diagnostics of token selection rather than as direct evidence of chemical causality.

3. Methodology

3.1. Problem formulation

Let a sample be (G, T, y), where $G = (V, E)$ is a periodic crystal graph, $T = (w_1, \ldots, w_L)$ is the Robocrystallographer description, and $y \in \mathbb{R}$ is a scalar property. The goal is to learn $f(G, T)$ minimizing mean absolute error (MAE). We construct two predictors, f_g and f_t, that use the same material input but different textual abstraction levels, and then combine their predictions at the output level.

3.2. Global and token-level prediction framework

Figure 1 traces the complete data flow without implying a single joint latent-fusion network. Formally, $\hat{y}_g = f_{\text{Global}}(G, T)$ and $\hat{y}_t = f_{\text{Token}}(G, T)$ are produced by independently trained models with checkpoints selected on the validation set. Each predictor contains its own SFTGNN instance. The two instances have the same architecture—92-dimensional atom inputs, five SFTConv layers, 128-dimensional node states, and mean pooling—but their parameters are not shared. Their text processing differs by abstraction level: the global pathway reads the MatBERT CLS state of the tokenized description available within the maximum sequence length, whereas the token-level pathway retains eligible non-special contextual token states and applies graph-conditioned attention. The final prediction-level mean is applied only to sample-aligned outputs. The comparison between the individual Global and Token predictors does not isolate textual granularity as a single causal factor, because their prediction heads and training objectives also differ: the Global path uses direct graph–CLS regression, whereas the Token path uses graph-conditioned token aggregation and an auxiliary graph loss.

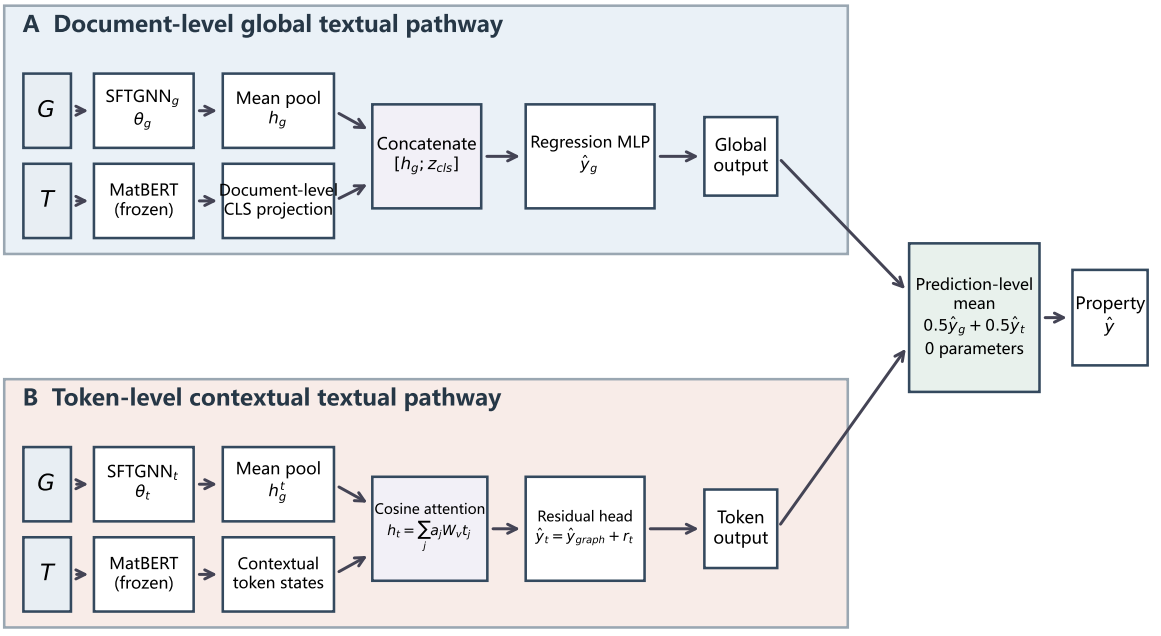


Figure 1: Global and token-level graph–text learning framework. A global textual pathway uses the document-level MatBERT CLS representation, whereas a token-level pathway uses graph-conditioned cosine attention over contextual token states. The pathways are trained independently under matched graph-encoder architectures, enabling prediction-level complementarity to be evaluated without a shared latent interaction module. Message-, latent-, and shared-backbone interaction variants are analyzed separately as diagnostic studies because their behavior is property dependent.

3.3. Spherical-harmonic graph encoder

For edge (j, i), the implementation computes a radial feature from reciprocal distance, $e_{ij}^r = \mathrm{MLP}_r(\mathrm{RBF}(1/d_{ij})) \in \mathbb{R}^{64}$, and projects spherical harmonics to $e_{ij}^s \in \mathbb{R}^{64}$. With $h_i, h_j \in \mathbb{R}^{128}$, the formal SFTConv implementation is

$$u_{ij} = [h_i; h_j; e_{ij}^r; e_{ij}^s] \in \mathbb{R}^{384}, \tag{1}$$

$$m_{ij} = \sigma(\mathrm{MLP}_{gate}([h_i; h_j; e_{ij}^r])) \odot \mathrm{MLP}_{msg}(u_{ij}), \tag{2}$$

$$h_i' = \mathrm{SiLU}\left(h_i + \mathrm{BN}\left(\sum_{j \in \mathcal{N}(i)} m_{ij}\right)\right). \tag{3}$$

Both MLPs use a 512-dimensional hidden layer and return 128 features. Five updates are followed by mean scatter pooling.

3.4. Global textual pathway

The global pathway corresponds to the original SFTMAT-style baseline used in this study: the SFTGNN graph representation is concatenated directly with the projected MatBERT CLS representation and passed to the regression head.

The CLS state $z_{cls} \in \mathbb{R}^{768}$ is projected by $768 \to 128 \to 64$ with ReLU. The global pathway concatenates it with the graph representation h_g:

$$\hat{y}_g = \mathrm{MLP}_g([h_g; \phi_{cls}(z_{cls})]). \tag{4}$$

This pathway preserves a document-level semantic summary.

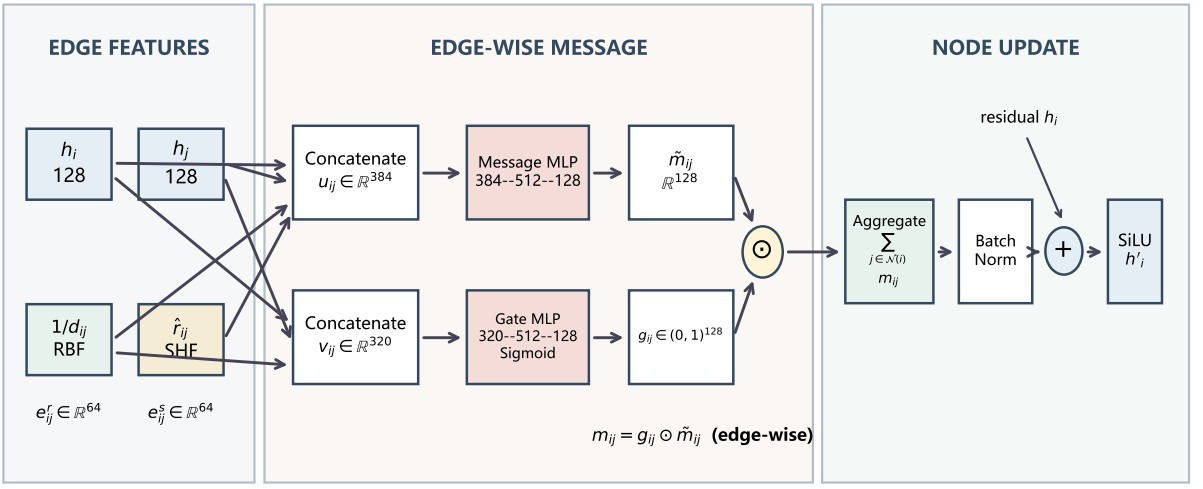


Figure 2: SFTConv message passing, matching Eq. (2): the Message MLP output and sigmoid Gate MLP output are multiplied element-wise for each edge before the gated messages are summed over neighbors. Batch normalization, residual addition, and SiLU follow aggregation.

3.5. Token-level textual pathway

MatBERT is frozen and cached at maximum length 128. The cache stores last hidden states with shape $[N, 128, 768]$, input IDs, and attention masks in BF16. Padding, CLS, and SEP positions are excluded from normalized attention. Retaining token identity allows graph-conditioned selection while keeping language-model computation outside the training loop.

Attention-score normalization. Raw attention uses $q = W_q h_g$ and $k_j = W_k t_j$. Because $q^\top k_j = \|q\|\|k_j\| \cos\theta_j$, scale and alignment are inseparable. Large norms amplify small angular differences and can make softmax nearly one-hot. We quantify sharpness by entropy $H(a) = -\sum_j a_j \log a_j$ and effective token count $N_{eff} = \exp(H(a))$.

Graph-conditioned aggregation. We normalize projected queries and keys and divide cosine similarity by a fixed temperature:

$$\bar{q} = \frac{W_q h_g}{\|W_q h_g\|_2}, \quad \bar{k}_j = \frac{W_k t_j}{\|W_k t_j\|_2}, \tag{5}$$

$$a_j = \frac{\exp(\bar{q}^\top \bar{k}_j / \tau)}{\sum_{m \in \mathcal{M}} \exp(\bar{q}^\top \bar{k}_m / \tau)}, \quad \tau = 0.5, \tag{6}$$

$$h_t = \mathrm{LayerNorm}\left(\sum_{j \in \mathcal{M}} a_j W_v t_j \right) \in \mathbb{R}^{128}. \tag{7}$$

$\mathcal{M}$ contains valid non-special tokens.

Residual prediction. The token-level predictor first produces a graph-only estimate and then a text-conditioned correction:

$$\hat{y}_{graph} = \psi_g(h_g), \qquad r_t = \psi_r([h_g; h_t]), \qquad \hat{y}_t = \hat{y}_{graph} + r_t. \tag{8}$$

Both heads use 128 or 256 hidden features with SiLU. The residual head allows the output to depend on both graph and aggregated text features; this parameterization alone does not establish a semantic interpretation of the correction.

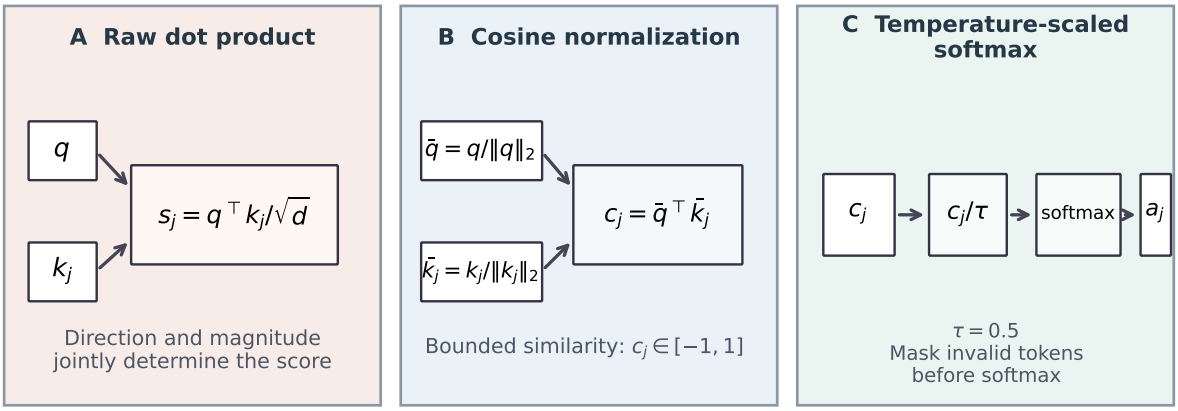


Figure 3: Mechanism of graph-conditioned token attention. Raw dot products combine vector direction and magnitude, cosine normalization bounds the similarity, and temperature-scaled softmax converts normalized scores into masked attention weights. The diagram contains no measured attention values.

3.6. Prediction-level combination

After independent training and checkpoint selection on the validation set, the sample-aligned predictions are combined as

$$\hat{y}_D = 0.5\hat{y}_g + 0.5\hat{y}_t. \tag{9}$$

The coefficient is fixed rather than fitted to validation or test labels, and the same rule is used for all ten tasks. This construction permits the contribution of prediction averaging to be assessed separately from the accuracy of either constituent predictor.

3.7. Training objective and computational complexity

The global textual pathway minimizes $\mathcal{L}_g = |\hat{y}_g - y|$. The token-level pathway uses $\mathcal{L}_t = |\hat{y}_t - y| + 0.2|\hat{y}_{graph} - y|$. Each pathway is trained independently, and its checkpoint is selected by minimum validation MAE. Token attention costs $O(BLd_a)$ beyond graph encoding, with $L = 128$ and $d_a = 128$. The combined prediction evaluates both predictors, trading additional latency and parameters for improved predictive accuracy.

4. Experiments

The evaluation separates three types of evidence. The main ten-task, two-run test experiment measures the performance of complete Global and Token predictors and their fixed average. A single-seed, six-task validation ablation compares text aggregators under a common head and objective. Same-view ensembles and error diagnostics help interpret the main gains, while interaction variants with unequal coverage provide exploratory context. A descriptive same-view analysis of saved test predictions is reported separately in Appendix A; it was not used for model or weight selection.

4.1. Datasets and tasks

We evaluate six JARVIS and four Materials Project property-prediction tasks following the benchmark data preparation and fixed partitions adopted by CrysMMNet [8]. The underlying crystal structures and property targets originate from JARVIS and the Materials Project [1, 2]; Matbench provides broader context for standardized materials benchmarks [34]. The ten tasks cover electronic, energetic, and mechanical properties. For Materials Project bulk and shear moduli, the benchmark loader applies a $\log_{10}$ transformation to the raw GPa targets before training and evaluation; the corresponding MAEs are therefore reported in the transformed target space.

4.2. Text generation and caching

Each crystal is converted to a Robocrystallographer description and tokenized by MatBERT. We refer to the tokenized crystallographic description available within the maximum sequence length, rather than assuming that every generated description is fully retained. Sequences use maximum length 128 with truncation and padding masks. The global path reads position zero as the CLS representation; the token path reads the full last-hidden-state tensor and removes CLS, SEP, and PAD from attention eligibility. MatBERT representations are precomputed in BF16 and paired with input IDs and masks, so language-model forward computation is not repeated during property-model training or inference when the cache is available.

4.3. Training protocol

The main Global, Token, and Global+Token experiments were repeated with two independent model-training seeds (42 and 43), while all models used the same fixed data partitions. Architecture, hyperparameters, target transformation, MatBERT sequence length, temperature $\tau = 0.5$, checkpoint criterion, fixed 0.5/0.5 prediction-level mean, and test protocol were unchanged. For each run, the global textual and token-level textual predictors are initialized and trained independently with batch size 32, evaluation batch size 64, AdamW [35], learning rate and maximum OneCycleLR rate 10^{-3}, weight decay 10^{-4}, BF16 autocast, and a 300-epoch budget. Each predictor is selected by its minimum validation MAE, after which the test set is evaluated once. Predictions are aligned by sample ID and their targets are matched within floating-point precision before averaging; band-gap outputs are clipped to nonnegative values. The interaction-location studies were validation-only diagnostic experiments and used the task and run coverage reported in Section 5.5; they were not uniformly repeated under the two-run main-test protocol. To isolate textual representation from pathway-specific prediction heads and training objectives, we additionally performed a single-seed matched validation ablation on a balanced six-task subset spanning JARVIS and Materials Project electronic, energetic, and mechanical properties. The CLS, masked-mean, and cosine-attention variants used the same graph encoder, 128-dimensional text output, regression head, MAE objective, optimizer, scheduler, training budget, validation partition, and training seed (42), and differed only in how the MatBERT representation was aggregated. These results were used only for representation-level analysis and were not evaluated on the test set.

4.4. Evaluation

MAE is the principal metric because it is in the benchmark target space and limits the influence of extreme errors. Relative gain against the global textual pathway is $(\mathrm{MAE}_g - \mathrm{MAE}_{dual})/\mathrm{MAE}_g \times 100\%$. We also report absolute-error Pearson correlation, oracle MAE, and per-sample win fractions on the validation set. Checkpoints are selected using validation MAE. The equal-weight combination is fixed and is not fitted to validation or test labels; the test set is excluded from training, tuning, and checkpoint selection.

The two-run validation predictions further define same-view controls. Let G_1, G_2 denote the Global predictions and T_1, T_2 the Token predictions from seeds 42 and 43. We compute $GG = (G_1 + G_2)/2$ and $TT = (T_1 + T_2)/2$, and compare them with the run-matched global–token means $GT_1 = (G_1 + T_1)/2$ and $GT_2 = (G_2 + T_2)/2$. All four files for each task contain identical sample-ID sets and ordering; targets agree within a maximum absolute numerical difference of 2.39×10^{-7}. No prediction weight is fitted from validation labels.

5. Results

5.1. Prediction averaging on the ten-task benchmark

Table 1 reports test-set MAE as the mean and sample standard deviation over two independent runs. Across the ten tasks, the test partition contributes 24,671 sample-task evaluations per run; this is not a count of unique crystals. The global–token prediction-level combination improves upon the global pathway on all ten tasks in both runs, with an average task-wise relative MAE reduction of approximately 5.9% (mean $\pm$ sample standard deviation over two independent runs: $5.875 \pm 0.036\%$). We use this small standard deviation only as evidence of consistency between the two runs, not as a formal significance estimate. In each run, the combined prediction also outperforms both individual pathways on nine of the ten tasks. One task per run is predicted slightly more accurately by the token-level pathway, while the combined prediction remains more accurate than the corresponding global pathway. The average macro relative gains over the global pathway are $5.747 \pm 0.069\%$ on JARVIS and $6.068 \pm 0.193\%$ on Materials Project.

Table 1: Test-set MAE averaged over two independent runs and relative gain over Global. Values are reported as mean ± sample standard deviation; lower MAE is better. Bold indicates the best result and underlining indicates the second-best result under the same evaluation protocol. Ranking compares the mean MAE; the standard deviation is displayed as part of the same result.

Task	Global	Token	Global+Token	Gain (%)
JARVIS-MBJ	0.259641 ± 0.002825	0.267325 ± 0.009433	**0.251800 ± 0.001808**	3.011 ± 1.752
JARVIS-OPT	0.124654 ± 0.003559	0.119036 ± 0.004411	**0.115764 ± 0.000876**	7.104 ± 1.950
JARVIS-Formation	0.029453 ± 0.000692	0.028832 ± 0.000190	**0.027437 ± 0.000406**	6.836 ± 0.810
JARVIS-Total	0.032296 ± 0.000405	0.031422 ± 0.000010	**0.029796 ± 0.000167**	7.736 ± 0.639
JARVIS-Bulk	9.60549 ± 0.14011	9.40847 ± 0.09234	**9.14175 ± 0.04797**	4.821 ± 0.889
JARVIS-Shear	8.96774 ± 0.03243	8.72852 ± 0.12811	**8.52204 ± 0.13611**	4.972 ± 1.174
MP-Bandgap	0.211599 ± 0.003322	**0.196220 ± 0.002127**	0.196354 ± 0.002946	7.204 ± 0.065
MP-Formation	0.024373 ± 0.000796	0.024256 ± 0.000511	**0.022769 ± 0.000646**	6.576 ± 0.399
MP-Bulk	0.037610 ± 0.001028	0.036156 ± 0.000402	**0.034938 ± 0.000623**	7.092 ± 0.884
MP-Shear	0.071981 ± 0.000318	0.071636 ± 0.000668	**0.069533 ± 0.000013**	3.400 ± 0.445
Macro gain				5.875 ± 0.036

The run-wise macro relative MAE reductions are 5.850% and 5.901%, showing close agreement in the overall benefit. Based on the task-wise means over the two runs, the combined prediction reduces MAE by 3.011–7.204% for electronic properties, 6.576–7.736% for energetic properties, and 3.400–7.092% for mechanical properties.

The improvement establishes the behavior of the combined configuration relative to its constituent predictors. It does not yet distinguish a cross-view benefit from ordinary ensembling; the next section addresses that comparison. Published benchmark values are collected in Appendix E for context and do not determine the principal conclusions.

5.2. Same-view and cross-view ensemble controls

The main result motivates a stronger comparison: whether the cross-view mean provides a benefit beyond averaging two predictors of the same type. Table 2 compares GG, TT, and GT using the paired validation predictions defined in Section 4.4.

Table 2: Validation performance of individual predictors and same- and cross-view prediction ensembles. Lower MAE is better; averaging is defined separately for each configuration below. Global and Token denote the mean MAE of the two independent runs. Global+Global and Token+Token average predictions across the two runs of the same pathway, whereas Global+Token denotes the mean MAE of the two run-matched cross-view averages. Bold indicates the best result and underlining indicates the second-best result under this common validation protocol. The normalized macro is computed task-wise relative to Global.

Task	Global	Token	Global+Global	Token+Token	Global+Token
JARVIS-MBJ	0.238892	0.247418	**0.230477**	0.235945	0.232514
JARVIS-OPT	0.123269	0.123004	**0.117583**	0.118742	0.117690
JARVIS-Formation	0.029294	0.028961	0.027565	**0.027346**	0.027449
JARVIS-Total	0.030939	0.030464	0.028922	**0.028319**	0.028642
JARVIS-Bulk	9.113950	9.128360	8.802365	8.806866	**8.798694**
JARVIS-Shear	8.652951	8.577900	8.397938	**8.335904**	8.343903
MP-Bandgap	0.213246	0.209270	0.202748	**0.199835**	0.201250
MP-Formation	0.020522	0.019973	0.019189	**0.018730**	0.018892
MP-Bulk	0.037887	0.036314	0.036431	**0.035190**	0.035667
MP-Shear	0.066456	0.064897	0.064436	0.063536	**0.063134**
10-task normalized macro	1.0000	–	0.9548	**0.9464**	0.9476

The cross-view average is lower than both constituent predictors at the level of the two-run mean validation MAE on all ten tasks. Same-view ensembles also improve prediction: GG has a normalized macro MAE of 0.9548 and TT

reaches 0.9464, compared with 0.9476 for GT. Thus, GT improves over repeating the Global pathway at the macro level, but does not improve over repeating the Token pathway. The observed gains cannot be attributed uniquely to combining two textual abstraction levels.

A post-hoc descriptive analysis of saved test predictions gives the same macro-level ordering between TT and GT: their normalized values are 0.935789 and 0.941171, respectively, and GT is lower on three of ten tasks (Appendix A). This analysis did not select a model or fusion weight. With only one cross-seed pair for each same-view ensemble, it does not establish statistical superiority of TT. It does show why improvement over a single Global baseline is insufficient evidence for an advantage of cross-view ensembling.

These controls concern complete predictor configurations. The following matched ablation asks a separate question: whether the text aggregator itself improves single-predictor accuracy under a common head and objective.

5.3. Matched text-representation ablation

To isolate the effect of textual representation, we compared three aggregation choices under a matched validation protocol. The balanced subset contains one task from each combination of dataset (JARVIS or Materials Project) and property family (electronic, energetic, or mechanical): Jarvis-Bandgap_MBJ, Jarvis-FormationEnergy, Jarvis-BulkModulusKv, MP-Bandgap, MP-FormationEnergy, and MP-ShearModuli. All three variants use the same SFT-GNN graph encoder, 128-dimensional text output, downstream regression head, MAE objective, optimizer, scheduler, training budget, validation split, and seed 42. They differ only in whether the MatBERT sequence is represented by its CLS state, a masked mean over eligible contextual tokens, or graph-conditioned cosine attention.

Table 3: Matched validation ablation of textual representation on the balanced six-task subset. Lower MAE is better. Bold indicates the best result and underlining the second-best result under the matched validation protocol. The normalized macro is computed task-wise relative to CLS and is not an average of raw MAEs.

Task	CLS	Masked Mean	Cosine Attention
JARVIS-MBJ	0.246605	**0.236364**	0.260371
JARVIS-Bulk	**8.971401**	9.361474	9.061776
JARVIS-Formation	**0.029189**	0.029192	0.029690
MP-Bandgap	0.217180	0.216474	**0.214222**
MP-Formation	0.020157	**0.019904**	0.020422
MP-Shear	**0.069412**	0.069488	0.075792
Normalized macro	1.0000	**0.9979**	1.0291

CLS, masked mean pooling, and cosine attention achieve the lowest MAE on 3, 2, and 1 tasks, respectively. Their normalized macro MAEs are 1.0000, 0.9979, and 1.0291. Cosine attention is therefore approximately 2.91% worse than CLS on this normalized macro, while masked mean pooling and CLS have similar aggregate values. This single-seed validation experiment provides no overall accuracy advantage for cosine attention over either simpler aggregator.

This result addresses aggregation within a matched predictor, not the source of complementarity in the main Global/Token pair. The main Token pathway also uses a different prediction head and auxiliary loss. Nor does the ablation establish which of the three matched predictors would contribute most to an ensemble: that would require their own paired ensemble comparisons. Single-predictor accuracy and ensemble utility must therefore remain distinct interpretations.

Cosine normalization removes query/key norm magnitude from the similarity score and leaves temperature τ as the explicit sharpness control. Available validation diagnostics show distributed rather than single-token attention, but these statistics characterize attention behavior only and do not establish universal predictive benefit or chemical causality.

5.4. Sample-level prediction-error diagnostics

To describe how the independently trained predictors differ, we examine paired absolute errors and a target-informed oracle on the validation set. These diagnostics characterize prediction-level complementarity; they do not identify whether its source is textual aggregation, head design, or training randomness.

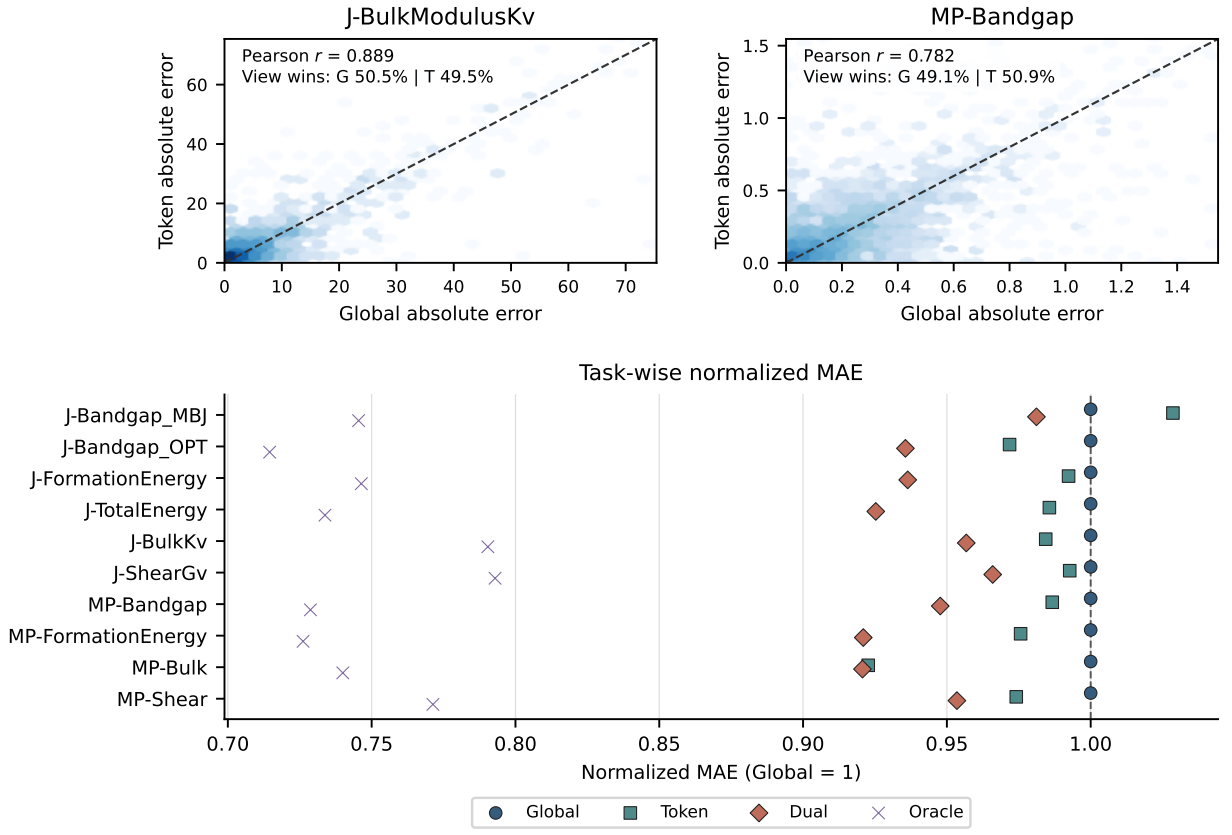


Figure 4: Representative sample-level error complementarity analysis from one independent run. Top: hexbin densities for two representative tasks with $y = x$, Pearson r, and Global/Token pathway-win ratios (ties excluded). Bottom: grouped dots for Global, Token, fixed prediction-level combination, and diagnostic Oracle MAE across all ten named tasks, normalized by the global pathway. Oracle uses targets and is non-deployable. A representative run is visualized for clarity; quantitative ranges for both runs are reported in Table 4.

Table 4: Run-wise ranges of validation-set complementarity diagnostics across the ten tasks. Win ratios include ties in the denominator.

Diagnostic	Run 1 range	Run 2 range
Pearson absolute-error correlation	0.614–0.900	0.591–0.906
Oracle gain over global pathway (%)	20.709–28.546	20.978–28.801
Global sample-win ratio (%)	17.732–48.526	15.147–48.780
Token sample-win ratio (%)	22.565–81.551	16.960–52.316
Combined gain over global pathway (%)	1.886–7.935	2.537–7.971

Across the ten tasks, Pearson absolute-error correlations span 0.614–0.900 in Run 1 and 0.591–0.906 in Run 2. The positive correlations are compatible with shared prediction difficulty, while values below one indicate non-identical absolute-error patterns. The oracle MAE is 20.709–28.546% lower than the global pathway in Run 1 and 20.978–28.801% lower in Run 2. Although oracle selection is not deployable because it uses the target, the persistent gap quantifies available sample-wise complementarity.

Together with the same-view controls, these diagnostics support complementary prediction errors in the evaluated models without assigning that complementarity uniquely to textual granularity.

5.5. Exploratory interaction diagnostics

To examine whether earlier cross-view interaction is beneficial, we evaluated validation-only latent-level, message-level, and shared-backbone variants. Because their task coverage, reference baselines, and computational budgets are not uniform, these experiments are interpreted as exploratory diagnostics rather than as a controlled ranking of fusion locations.

Latent interaction through query conditioning, feature refinement, and explicit pairwise coupling produced measurable representation changes, but predictive effects varied across tasks and datasets. Similarly, SEC-SFT text-conditioned message modulation improved several JARVIS tasks while degrading part of the Materials Project mechanical subset; a zero-initialized residual formulation repaired some degradations but did not preserve gains uniformly.

Shared-backbone dual-view fusion retained some sample-wise complementarity: on the balanced six-task subset, the fixed mean improved four tasks and was lower than the better shared branch on five. Taken together, earlier interaction is feasible but remains property dependent under the available validation protocols. Detailed results appear in Appendices B–D. These studies supplement the main ensemble and aggregation analyses but do not establish why one fusion location should be preferred.

5.6. Computational cost

The Global and Token predictors contain 2,661,154 and 2,800,739 trainable parameters, respectively, giving 5,461,893 parameters when both independently trained predictors are retained. The fixed prediction mean introduces no additional trainable parameters, although inference requires both graph–text pathways. The Global pathway uses a CLS representation, whereas the Token pathway retains cached contextual token states.

6. Discussion

The experiments distinguish three observations that are easily conflated in graph–text prediction: an ensemble can outperform its constituent predictors, a particular text aggregator can improve a matched single predictor, and a cross-view ensemble can outperform same-view averaging. The present evidence supports the first observation, but does not establish the latter two as general advantages of the proposed Token pathway.

Prediction complementarity. The fixed Global–Token mean improves over Global on all ten test tasks in both training runs and over both constituent predictors on nine tasks per run. The paired errors and oracle gaps are consistent with complementary predictions. These observations concern the trained predictors, whose aggregation mechanisms, heads, objectives, and learned parameters differ. They do not by themselves demonstrate complementary semantic information in the two text representations.

Aggregation and ensemble controls. Under a common head and loss, cosine attention has a higher normalized macro MAE than CLS and masked mean pooling on the six-task validation subset. Separately, TT has a lower normalized macro MAE than GT in the validation controls and the descriptive test analysis. Taken together, these findings rule out interpreting the main gain as evidence that finer text aggregation is inherently more accurate or that cross-view averaging necessarily exceeds same-view averaging. They do not quantify the causal share of the gain due to each architectural or training difference.

Scope of the interaction studies. Earlier latent-, message-, and shared-backbone interactions produced property-dependent outcomes under nonuniform protocols. Their role is to document additional behavior, not to justify a ranking against independent prediction averaging. The latter remains the configuration evaluated in the main ten-task test experiment; its performance should be interpreted within that protocol.

For materials-property prediction, the practical implication is to evaluate text aggregation and ensemble composition separately for the target properties. A more detailed text representation or a measurable interaction signal is not sufficient evidence of an accuracy gain. Reporting same-view controls alongside cross-view results makes that distinction visible, including cases in which a simpler aggregation or a same-view ensemble performs as well or better.

7. Limitations

The current evaluation uses two independent runs, supporting consistency but not formal significance testing; same-view controls also show that ordinary averaging contributes materially. The main Global/Token comparison jointly changes textual abstraction, prediction head, and training objective, while the matched CLS/mean/cosine ablation partially isolates representation choice on one seed and six validation tasks, not as a full two-run replacement.

Published benchmarks use different partitions and target transformations. The two-predictor design increases storage and inference cost, and cache generation, wall-clock time, latency, and peak memory were not benchmarked under one hardware protocol. Robocrystallographer descriptions inherit the scope and heuristics of the generation procedure, and other intermediate text abstractions remain unexplored.

The interaction studies are validation-only, have unequal task coverage, and were not used for test-time model selection. Aggregate attention statistics do not establish chemical causality or sample-level interpretability, and no matched raw-dot-product versus cosine-normalized performance ablation is available; normalization is therefore treated as a numerical control rather than a causal performance claim.

8. Conclusion

This study examined text aggregation and prediction averaging in SFTGNN–MatBERT crystal-property predictors. Across six JARVIS and four Materials Project tasks with fixed partitions and two training seeds, a fixed 0.5/0.5 Global–Token mean improved over Global on every task and over both constituent predictors on nine tasks in each run. The average task-wise relative MAE reduction against Global was approximately 5.9% ($5.875 \pm 0.036\%$ across the two runs).

The controls delimit the meaning of this gain. In a matched six-task validation ablation, cosine attention did not improve aggregate accuracy over CLS or masked mean pooling. Same-view ensembles also improved performance, and TT had a lower normalized macro MAE than GT. The supported finding is therefore complementarity between the evaluated predictors, rather than a demonstrated advantage of textual granularity or cross-view ensembling over the strongest same-view control. Exploratory interaction results further describe property dependence without establishing a preferred fusion location. These comparisons emphasize the need to assess aggregation accuracy, ensemble gains, and their possible sources as distinct empirical questions.

A. Post-hoc descriptive same-view analysis

This post-hoc descriptive analysis uses saved test predictions and does not select an architecture or fusion weight. For each task, GG and TT average predictions across the two runs of the corresponding pathway. The GT value is the mean of the two MAEs obtained from run-matched Global–Token prediction averages; it is not a four-model prediction average. Each task MAE is divided by the mean of the two Global MAEs before averaging across the ten tasks. The normalized macro values are Global = 1.000000, GG = 0.956587, TT = 0.935789, and GT = 0.941171; GT is lower than GG on 9/10 tasks but lower than TT on 3/10. The resulting 5.8829% versus 5.8754% reductions differ only because of aggregation order.

B. Latent interaction diagnostics

These validation-only variants were designed to probe distinct latent interaction mechanisms and were not candidate final models. TAGIF denotes token-anchored global interaction fusion; GCTA denotes CLS-conditioned graph–token attention; GCTR denotes CLS-conditioned token refinement; and PVIN denotes a parallel-view interaction network. Table 5 summarizes their representative diagnostics, including cases in which the latent interaction was active but its predictive effect was weak or property dependent.

Table 5: Validation-only diagnostics for latent interaction variants. All values are computed on validation sets; no test evaluation is used for these variants.

Method	Task	Predictive result	Representation or attention diagnostic	Semantic/control evidence
TAGIF	JARVIS-Bandgap_MBJ	−0.975% macro gain vs Token-only over two tasks	innovation ratio 0.989; interaction ratio 0.432	correct CLS control better than shuffled/zero in both tasks
TAGIF	MP-Bandgap	0/2 wins vs Token-only	innovation ratio 0.834; interaction ratio 0.695	non-collapsed orthogonal component
GCTA	JARVIS-Bandgap_MBJ	valid MAE 0.285429 vs 0.286644 for naive concat	$\|\|q_c\|\|/\|\|q_g\|\| = 0.440$; attention L1 change 0.00218	active W_{qc} gradient
GCTA	MP-Bandgap	valid MAE 0.217874 vs 0.219016 for naive concat	$\|\|q_c\|\|/\|\|q_g\|\| = 0.194$; attention L1 change 0.00161	active W_{qc} gradient
GCTR	JARVIS-Bandgap_MBJ	degrades by 0.758% vs naive concat	$\cos(h_t, h_{refined}) = 0.968$; refinement ratio 0.028	FiLM branch active
GCTR	MP-Bandgap	improves by 1.835% vs naive concat	$\cos(h_t, h_{refined}) = 0.856$; refinement ratio 0.016	semantic control not consistently effective

C. Message-level interaction diagnostics

SEC-SFT introduces text-conditioned modulation into the final SFTConv message construction. The residual formulation adds a zero-initialized text-conditioned correction to the original message. The following tables preserve task-level validation values under the task-specific coverage described in Section 5.5.

Table 6: Message-level validation diagnostics. Panel (a) reports task-wise SEC-SFT relative changes; panel (b) reports the selected residual SEC-SFT comparisons. All values are validation results.

(a) SEC-SFT task-wise relative change

Task	Relative change (%)
JARVIS-MBJ	+1.283
JARVIS-OPT	+0.660
JARVIS-Formation	-1.121
JARVIS-Total	+1.740
JARVIS-Bulk	+2.418
JARVIS-Shear	+2.279
MP-Bandgap	+2.206
MP-Formation	+1.675
MP-Bulk	-3.844
MP-Shear	-5.553

(b) Residual SEC-SFT selected cases

Task	Original	SEC-SFT	Residual SEC-SFT
MP-Bulk	0.03789499	0.03935185	0.03717689
MP-Shear	0.06582852	0.06948390	0.06564737
JARVIS-Bulk	9.2148	8.9920	9.3033

D. Shared-backbone prediction diagnostics

Shared-backbone dual-view fusion (SBDVF) uses two textual prediction heads with a shared SFTGNN encoder. The following tables report detailed validation diagnostics for the balanced six-task subset.

Table 7: Detailed shared-backbone validation diagnostics for the balanced six-task subset.

Task	Abs.-error corr.	Oracle gain (%)	Gradient cosine
JARVIS-MBJ	0.941	13.010	+0.477
JARVIS-Formation	0.988	9.381	-0.435
JARVIS-Bulk	0.987	6.273	+0.956
MP-Bandgap	0.929	11.582	+0.476
MP-Formation	0.981	10.755	-0.215
MP-Shear	0.958	10.140	+0.690

Table 8: Shared-backbone dual-view validation diagnostics for the balanced six-task subset. Values are MAE; the gain is relative to the matched validation baseline.

Task	Original	Shared Global	Shared Token	Final	Gain (%)
JARVIS-MBJ	0.240074	0.247644	0.244046	0.241591	-0.632
JARVIS-Formation	0.028966	0.028823	0.028645	0.028505	+1.591
JARVIS-Bulk	9.214838	8.939804	8.887833	8.869501	+3.748
MP-Bandgap	0.212996	0.204922	0.202905	0.200832	+5.711
MP-Formation	0.020460	0.020200	0.020050	0.019905	+2.712
MP-Shear	0.065829	0.066835	0.068522	0.066976	-1.744
Macro gain					+1.898

E. Published benchmark context

Because the published values use heterogeneous data partitions, target transformations, and evaluation protocols, Tables 9 and 10 are provided only as literature context. All principal conclusions in this work are based on internally reproduced, sample-aligned comparisons.

Table 9: Reference comparison under different protocols on JARVIS. MAE units are eV/atom for formation and total energy, eV for band gaps, and GPa for elastic moduli. Published values follow the evaluation protocol of each study; dashes indicate unreported properties. Data partitions, target transformations, and test protocols differ across studies, so the values do not establish a protocol-matched ranking.

Method	Formation energy	Bandgap OPT	Bandgap MBJ	Total energy	Bulk K_v	Shear G_v
CGCNN	0.063	0.200	0.413	0.078	14.47	11.75
SchNet	0.045	0.192	0.433	0.047	14.33	10.67
MEGNet	0.047	0.145	0.344	0.058	15.11	13.09
GATGNN	0.047	0.170	0.513	0.056	14.32	12.48
ALIGNN	0.033	0.142	0.310	0.037	10.40	9.481
Matformer	0.0325	0.137	0.300	0.035	–	–
PotNet	0.0294	0.127	0.272	0.032	–	–
CrysMMNet	0.028	0.128	0.278	0.034	9.625	8.471
eComFormer	0.0284	0.124	0.280	0.032	–	–
iComFormer	0.0272	0.122	0.260	0.0288	–	–
SFTGNN	0.0283	0.129	0.270	0.031	–	–
Ours (Global+Token)	0.027437	0.115764	0.251800	0.029796	9.14175	8.52204

Table 10: Reference comparison under different protocols on Materials Project. MAE units are eV/atom for formation energy, eV for band gap, and $\log_{10}$(GPa) for elastic moduli. Published values follow the evaluation protocol of each study. Data partitions, target transformations, and test protocols differ across studies, so the values do not establish a protocol-matched ranking.

Method	Formation energy	Bandgap	Bulk moduli	Shear moduli
CGCNN	0.031	0.292	0.047	0.077
SchNet	0.033	0.345	0.066	0.099
MEGNet	0.030	0.307	0.060	0.099
GATGNN	0.033	0.280	0.045	0.075
ALIGNN	0.022	0.218	0.051	0.078
Matformer	0.021	0.211	0.043	0.073
PotNet	0.0188	0.204	0.040	0.0650
CrysMMNet	0.020	0.197	0.038	0.062
eComFormer	0.01816	0.202	0.0417	0.0729
iComFormer	0.01826	0.193	0.0380	0.0637
SFTGNN	0.0183	0.198	0.037	0.064
Ours (Global+Token)	0.022769	0.196354	0.034938	0.069533

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Data availability

The source data used in this study are publicly available from JARVIS and Materials Project. Derived descriptions and prediction files supporting the findings will be made available with the final release.

Code availability

The model and analysis code are available at https://github.com/luoai4430-ship-it/global-token-crystal-property-prediction.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used OpenAI Codex to assist with writing and organizing code files and with language editing and manuscript organization. The authors reviewed and verified the generated code and take full responsibility for the research design, data processing, experiments, analyses, and manuscript content.

Declaration of competing interest

The authors declare no competing interests.

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